WORKED SOLUTION

STEP II 2020, Question 1

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Each integrand is a product of powers of two linear factors whose exponents add up to -2. For such integrands the right substitution is the *ratio* of the two linear factors (or its square root): it absorbs both brackets at once, and what is left is a simple power or a standard arctangent. Part (i) shows this happening with the given substitution, part (ii) asks you to copy the trick, and part (iii) asks you to spot it for yourself.

Solution

Part (i)

With $x = \dfrac{1}{1-u}$ we have

$$x - 1 = \frac{1-(1-u)}{1-u} = \frac{u}{1-u}, \qquad \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{(1-u)^2}.$$

As u runs over $(0,1)$, x runs over $(1,\infty)$, and $u = 1 - \frac{1}{x} = \frac{x-1}{x}$. All the quantities under square roots are positive because $0 < u < 1$, so

$$x^{3/2}(x-1)^{1/2} = \frac{1}{(1-u)^{3/2}} \cdot \frac{u^{1/2}}{(1-u)^{1/2}} = \frac{u^{1/2}}{(1-u)^2}.$$

The factors of $(1-u)^2$ now cancel exactly against $\mathrm{d}x/\mathrm{d}u$:

$$\begin{aligned}\int \frac{1}{x^{3/2}(x-1)^{1/2}}\,\mathrm{d}x &= \int \frac{(1-u)^2}{u^{1/2}} \cdot \frac{1}{(1-u)^2}\,\mathrm{d}u = \int u^{-1/2}\,\mathrm{d}u \\ &= 2u^{1/2} + c = 2\sqrt{\frac{x-1}{x}} + c.\end{aligned}$$

Part (ii)

In part (i), $u = \frac{x-1}{x}$ was the ratio of the two linear factors. Here we copy that idea and take

$$u = \frac{x-2}{x+1}, \qquad \text{equivalently} \qquad x+1 = \frac{3}{1-u}, \quad x-2 = \frac{3u}{1-u},$$

since $x-2 = (x+1)-3$. For $x > 2$ we get $0 < u < 1$, and $\dfrac{\mathrm{d}x}{\mathrm{d}u} = \dfrac{3}{(1-u)^2}$. Then

$$(x-2)^{3/2}(x+1)^{1/2} = \frac{3^{3/2}u^{3/2}}{(1-u)^{3/2}} \cdot \frac{3^{1/2}}{(1-u)^{1/2}} = \frac{9u^{3/2}}{(1-u)^2},$$

and so

$$\begin{aligned}\int \frac{1}{(x-2)^{3/2}(x+1)^{1/2}}\,\mathrm{d}x &= \int \frac{(1-u)^2}{9u^{3/2}} \cdot \frac{3}{(1-u)^2}\,\mathrm{d}u = \frac{1}{3}\int u^{-3/2}\,\mathrm{d}u \\ &= -\frac{2}{3}u^{-1/2} + c = -\frac{2}{3}\sqrt{\frac{x+1}{x-2}} + c.\end{aligned}$$

Differentiating confirms it. With $f = \frac{x+1}{x-2}$ we have $f' = -\frac{3}{(x-2)^2}$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(-\tfrac{2}{3}\sqrt{f}\right) = -\frac{f'}{3\sqrt{f}} = \frac{1}{(x-2)^2}\sqrt{\frac{x-2}{x+1}} = \frac{1}{(x-2)^{3/2}(x+1)^{1/2}}.$$

Part (iii)

This time no substitution is given. The integrand is not a pure power of a ratio because of the extra factor $x-1$, but the two square roots are of the same type as before.

> **Ask yourself** Both earlier answers were square roots of a ratio of the linear factors. What happens if we make such a square root the new variable?

Try

$$t = \sqrt{\frac{x-2}{3x-2}}.$$

As x increases from 2 to ∞, $\frac{x-2}{3x-2} = \frac{1}{3} - \frac{4}{3(3x-2)}$ increases from 0 towards $\frac{1}{3}$, so t increases from 0 to $\frac{1}{\sqrt{3}}$ and $0 \le t < \frac{1}{\sqrt{3}}$ throughout. Solving $t^2(3x-2) = x-2$ for x gives

$$x = \frac{2(1-t^2)}{1-3t^2},$$

where $1-3t^2 > 0$. From this,

$$x-1 = \frac{1+t^2}{1-3t^2}, \qquad 3x-2 = \frac{4}{1-3t^2},$$

and

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{2[-2t(1-3t^2)+6t(1-t^2)]}{(1-3t^2)^2} = \frac{8t}{(1-3t^2)^2}.$$

The two square roots combine neatly, since $\sqrt{x-2} = t\sqrt{3x-2}$:

$$(x-2)^{1/2}(3x-2)^{1/2} = t\,(3x-2) = \frac{4t}{1-3t^2}.$$

So the whole denominator is

$$(x-1)(x-2)^{1/2}(3x-2)^{1/2} = \frac{4t(1+t^2)}{(1-3t^2)^2},$$

and once again everything cancels against $\mathrm{d}x/\mathrm{d}t$:

$$\int_2^\infty \frac{\mathrm{d}x}{(x-1)(x-2)^{1/2}(3x-2)^{1/2}} = \int_0^{1/\sqrt{3}} \frac{(1-3t^2)^2}{4t(1+t^2)} \cdot \frac{8t}{(1-3t^2)^2}\,\mathrm{d}t = \int_0^{1/\sqrt{3}} \frac{2}{1+t^2}\,\mathrm{d}t$$
$$= 2\Big[\arctan t\Big]_0^{1/\sqrt{3}} = 2 \cdot \frac{\pi}{6} = \frac{\pi}{3},$$

as required.

The original integral is improper at both ends, but it converges. Near $x = 2$ the integrand behaves like $\frac{1}{2}(x-2)^{-1/2}$, and for large x it behaves like $\frac{1}{\sqrt{3}}x^{-2}$. The substitution turns it into an ordinary integral of a continuous function over a finite interval, so no limits need to be taken separately.

Checks

- Differentiate the answers to (i) and (ii). For (i), $\frac{\mathrm{d}}{\mathrm{d}x} 2\sqrt{1-\frac{1}{x}} = \frac{1}{x^2}\left(1-\frac{1}{x}\right)^{-1/2} = x^{-3/2}(x-1)^{-1/2}$.
- Signs: the integrand in (ii) is positive, so the antiderivative must increase. It does, since $\sqrt{\frac{x+1}{x-2}}$ decreases towards 1 as x grows.
- Numerical: $\int_{1.5}^{3}$ in (i) and $\int_{2.5}^{4}$ in (ii), evaluated by Simpson's rule, agree with the antiderivatives (0.47829… and 0.70974…). The integral in (iii), computed directly by splitting at $x = 3$, gives $1.0471975\ldots = \frac{\pi}{3}$.

Common slips

- Forgetting to change the limits in (iii). The upper limit is $t = \frac{1}{\sqrt{3}}$, not ∞, which is where the $\frac{\pi}{6}$ comes from.
- Writing the answer to (i) or (ii) in terms of u and not converting back to x, as the question asks.
- Mishandling the half-powers of $1-u$. Collect them carefully: $(1-u)^{-3/2}(1-u)^{-1/2} = (1-u)^{-2}$.

Beyond STEP

Integrands built from $\sqrt{(x-a)(x-b)}$ can always be made rational by a substitution like ours. This is part of the classical theory of integrating rational functions of x and $\sqrt{\text{quadratic}}$ (Euler's substitutions). Part (iii) is also an improper integral over an infinite range, and making sense of these with limits is a standard first-year analysis topic. Abbott's *Understanding Analysis* treats improper integrals carefully, and Siklos's *Advanced Problems in Mathematics* has more STEP integrals of this kind.

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